MATH 201 Your name: _____

Learning target 2.3, version 1

Compute derivatives using the product and quotient rules.

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1. Consider the function $g(t) = 7 \cdot t^4$.

   (a) Compute $g'(t)$ using the product rule. Simplify as much as possible.

   (b) Now compute $g'(t)$ again, *without* the product rule.

   (c) Are both ways *correct*? Which way do you *prefer*?

2. Consider the function $f(t) = \dfrac{t^4}{7}$.

   (a) Compute $f'(t)$ using the quotient rule. Simplify as much as possible.

   (b) Now compute $f'(t)$ again, *without* the quotient rule.
   (Hint: What's another way to say "divide by 7"?)

   (c) Are both ways *correct*? Which way do you *prefer*?